

Why don't we cool AI data centres with salt water?

Posted by u/Vikster2468 • 0 points • 11 comments

One huge problem with AI data centres is that clean, drinkable water is used to cool them, which can strain water available for municipalities. If the water is just being used as a heat sink, why aren't we using salt water instead?

Comments

u/MrDavieT • 1 points • 2026-03-05 18:37 UTC

Corrosion

u/Bobbbob34 • 1 points • 2026-03-05 17:46 UTC

>One huge problem with AI data centres is that clean, drinkable water is used to cool them, which can strain water available for municipalities. If the water is just being used as a heat sink, why aren't we using salt water instead?

Because salt water destroys pretty much all it touches?

u/Long-Aardvark-3129 • 4 points • 2026-03-05 17:45 UTC

Same question asked 20 times by ... bots?

u/Vikster2468 • 1 points • 2026-03-05 18:26 UTC

I scrolled for a while and didn't see it asked, I guess I should have looked harder? I was watching a tiktok about the water issues with the data centres, and was wondering why when the answer seemed so simple (just use non-drinkable water to start with)... that's why I asked.

u/DiamondJim222 • 3 points • 2026-03-05 20:31 UTC

Among others, just asked earlier today:

https://www.reddit.com/r/NoStupidQuestions/comments/1rlpht/why_dont_data_centers_use_salt_water/

u/GyantSpyder • 2 points • 2026-03-05 17:43 UTC

Salt water is not a good coolant because the salt corrodes machinery and because if the water boils or evaporates it leaves behind salt deposits.

There's a difference between a coolant and a heat sink. Coolant is actively circulated so the heat is moved somewhere else, heat sinks are stationary and absorb a heat with the idea that they gradually cool off.

If water were just a heat sink, then salt water might work, because you wouldn't have to move it anywhere. But you do have to move it places, so the salt it carries around is a problem.

u/Vikster2468 • 1 points • 2026-03-05 18:22 UTC

Thank you so much!! That answer makes a lot of sense and I didn't think about how the water is cooling other than by just being there. I appreciate you taking the time!

u/[deleted] • 1 points • 2026-03-05 20:30 UTC

[deleted]

u/Moogatron88 • 3 points • 2026-03-05 17:41 UTC

From what I understand they are looking into this as an option. But it would require specialised infrastructure and would have serious issues with corrosion and such.

u/Feisty-Collar-9866 • 2 points • 2026-03-05 17:41 UTC

Salt water would corrode the cooling systems pretty fast so you'd be replacing equipment way more often. Plus most data centers aren't near ocean so you'd need to transport massive amounts of saltwater which gets expensive quick

The real issue is that cooling systems need specific water chemistry to work properly and salt just messes with everything

u/Jim777PS3 • 1 points • 2026-03-05 17:40 UTC

Salt water eats everything it touches. Its corrosive.

u/Pastadseven • 2 points • 2026-03-05 17:40 UTC

Saltwater is corrosive as fuck. And heavy. So you'd need to put the data center...on the coast. Which is famously easy to find industrial space on.